

Deep Learning for Leaf Image Sequence Alignment of Field Crops

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Introduction:

Motivation and Challenges

- **Motivation:**

- Tracking crop disease symptoms over time helps understand disease progression
- Important for studying effects of:
 - Genotype
 - Treatment
 - Environmental conditions
- Manual assessment is slow and subjective
- Registering leaf images from different time points simplifies measuring the growth and emergence of individual disease symptoms over time.

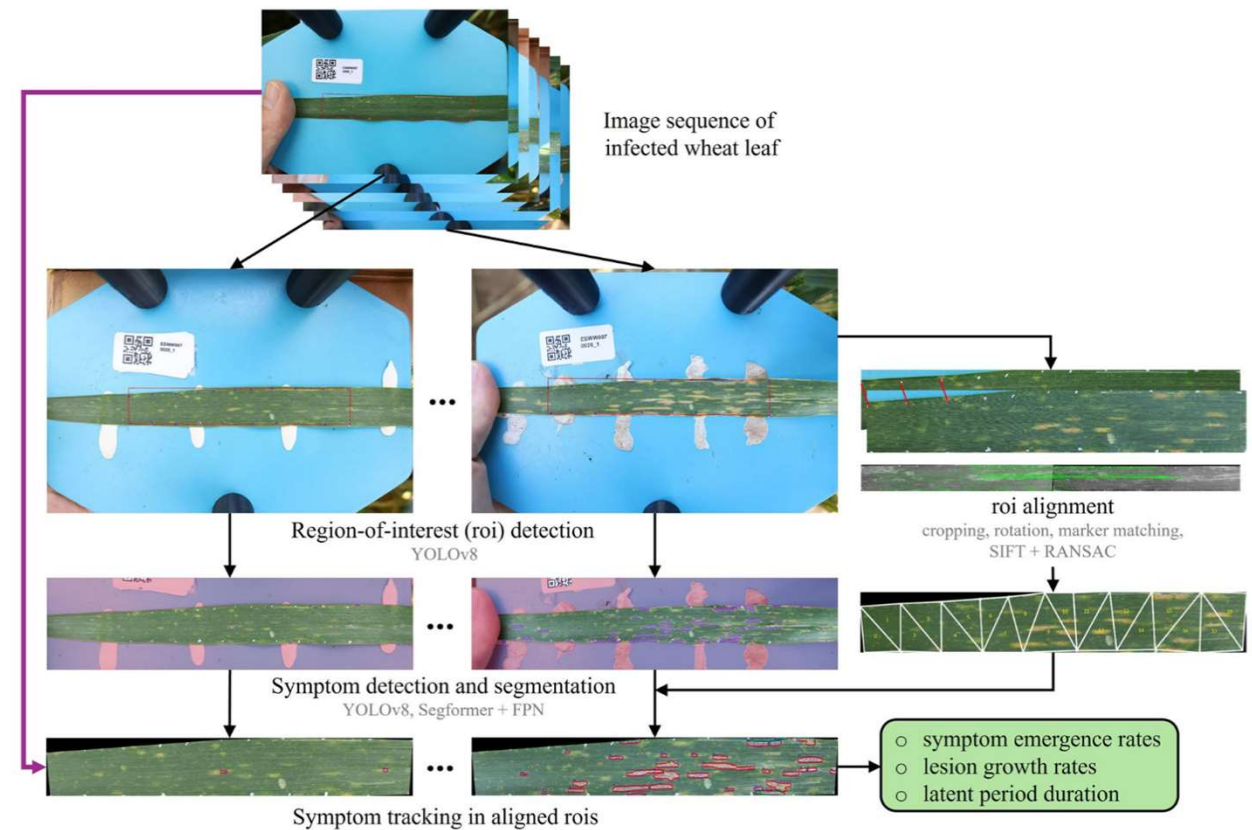
- **Challenges:**

- Leaves are not rigid $\Rightarrow$ local deformations
- Leaf appearance can change drastically over time
- Wheat leaves have low texture and no distinct features

Introduction:

Existing Approach – Piecewise Affine Registration

- Artificial markers placed along leaf edges
- Detect markers → match them across frames
→ subdivide leaf into triangles → apply affine transform for each triangle
- Good registration accuracy (manual evaluation)
- But:
 - Manual labor needed for placing markers
 - Proper matching of markers required time-consuming optimization of a series of parameters
 - Markers can be lost
 - Part of leaf area becomes unusable



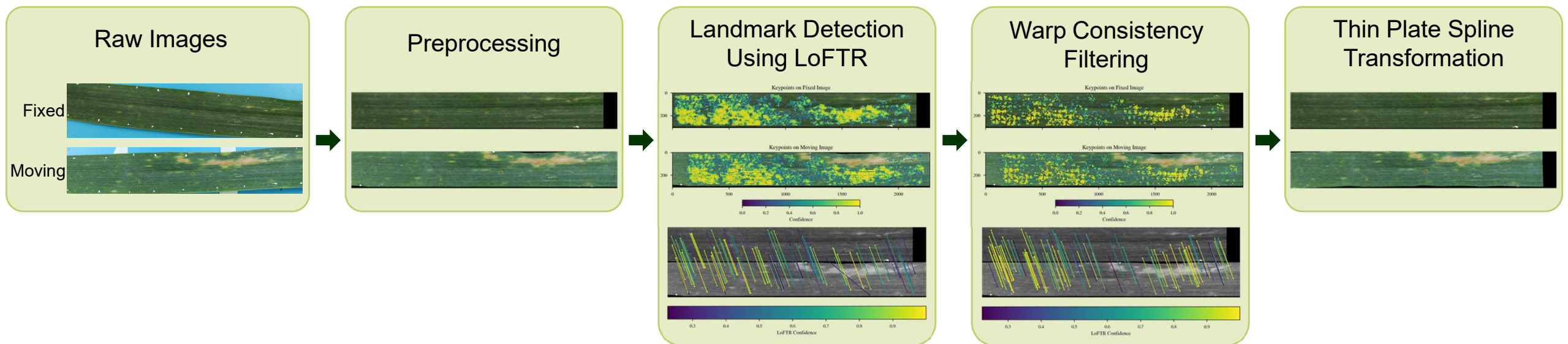
Source: [2]

For details, see [1].

Introduction:

Goal and Approach

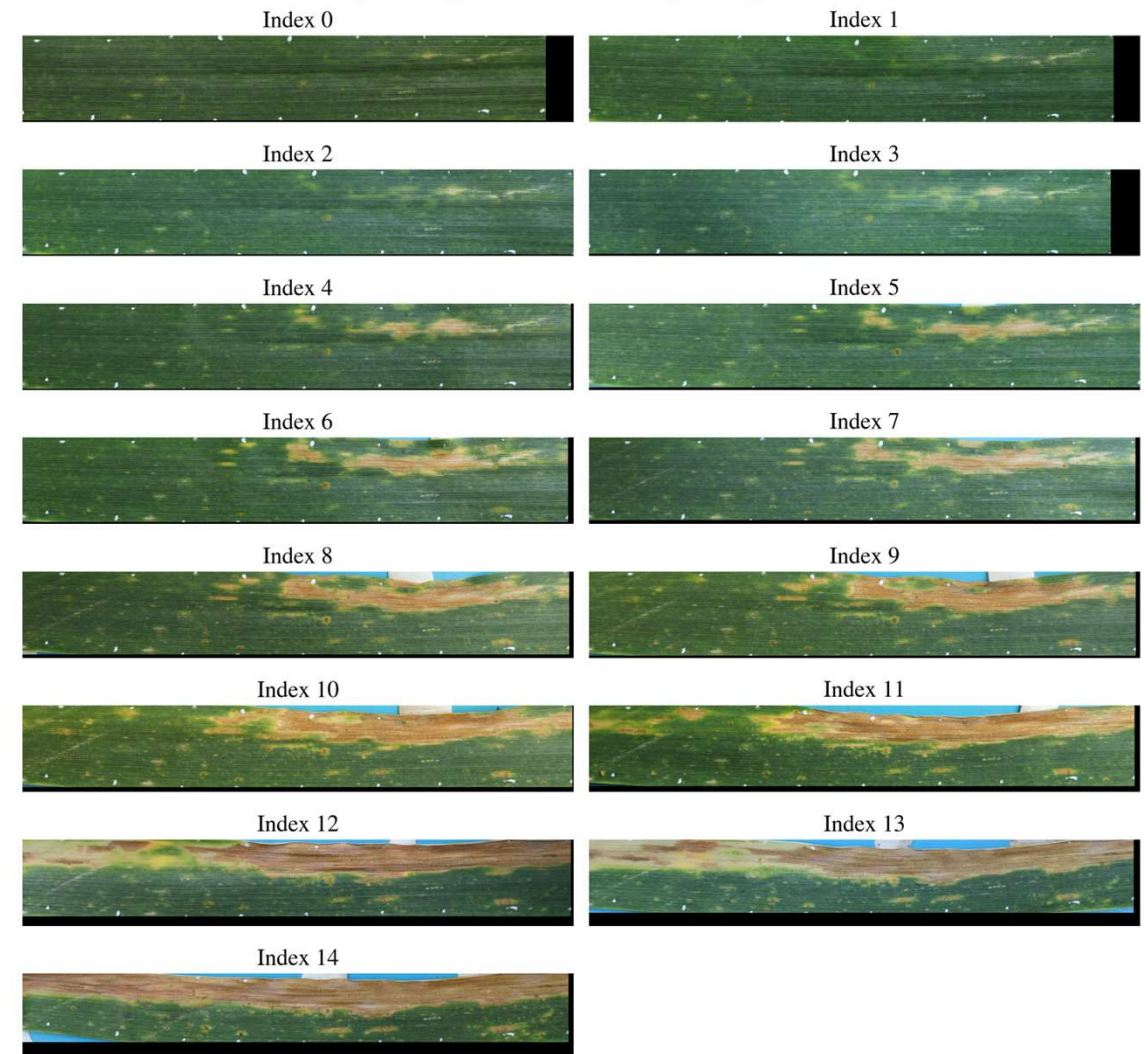
- **Goal:** Devise an image registration method that doesn't rely on artificial markers and is robust to drastic changes in appearance
- **Approach:** Use the local image feature matching method LoFTR to detect matches between image pairs, and use those matches as landmarks to fit a Thin-Plate-Spline transform between the images.



Dataset

- Collection of time series of high-resolution wheat leaf images with developing disease symptoms
- 1032 individual wheat leaves with on average 12 images at different time points
- On each leaf, 10-15 white dots were applied along both edges of the leaf
- Dataset provides:
 - Images of the full leaf
+ corresponding segmentation masks
 - Images of the Region of Interest only
+ corresponding segmentation masks
 - ROIs registered with Piecewise Affine approach
+ corresponding segmentation masks

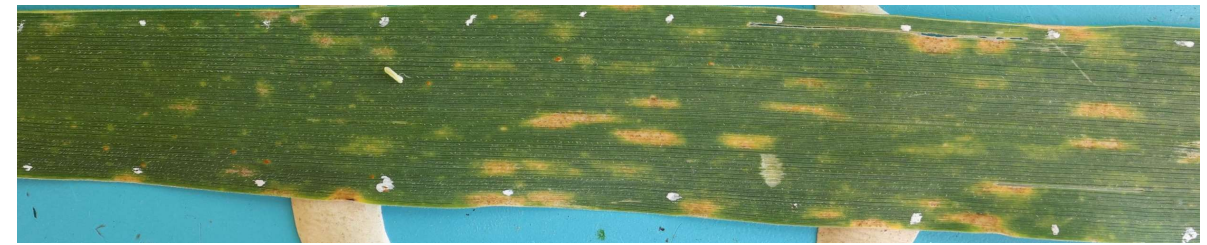
Fig: Sample of Leaf Image Sequence



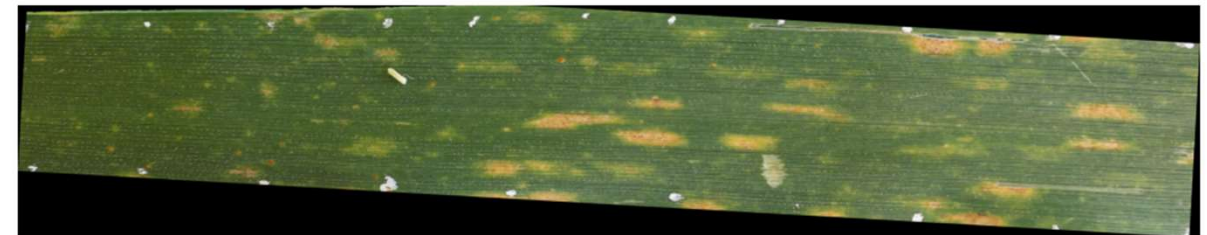
For details, see [3].

Method: Preprocessing

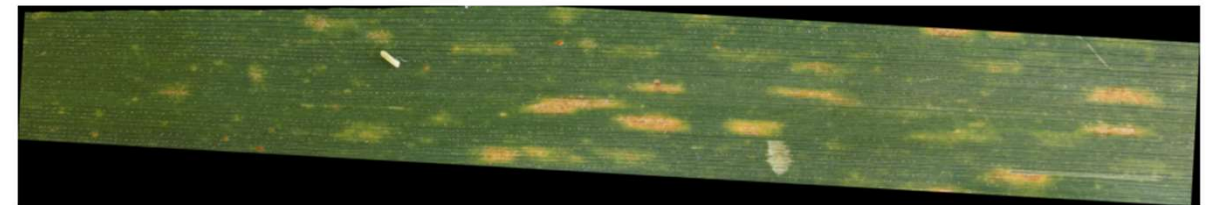
- Downscale images to avoid memory limitations
- Remove all superfluous background
- Mask out non-leaf background
- Erode mask to artificially remove markers
- Align ROI with image axes



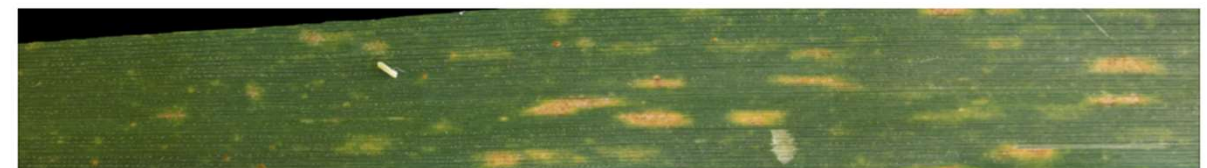
Mask background



Erode markers

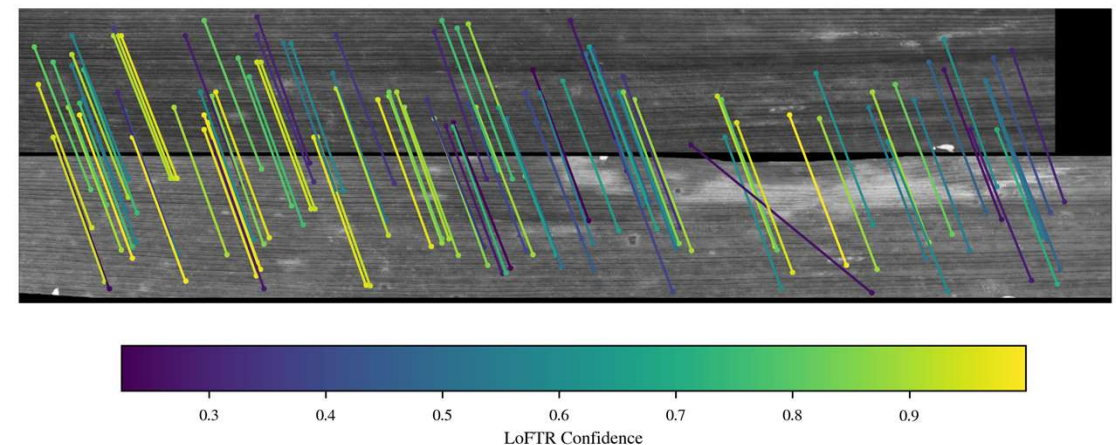
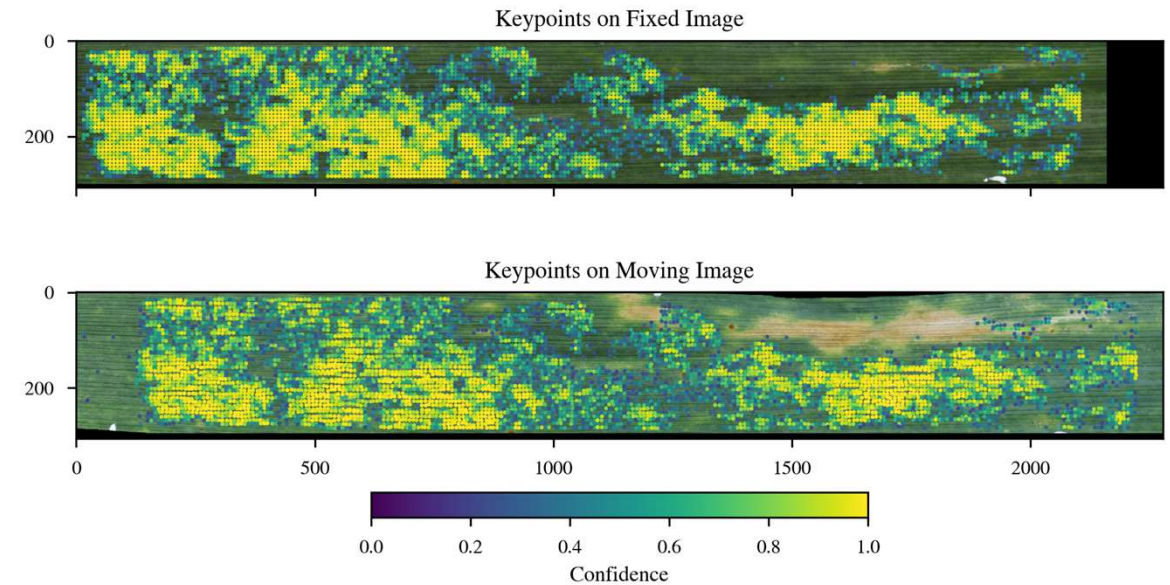


Rotate ROI

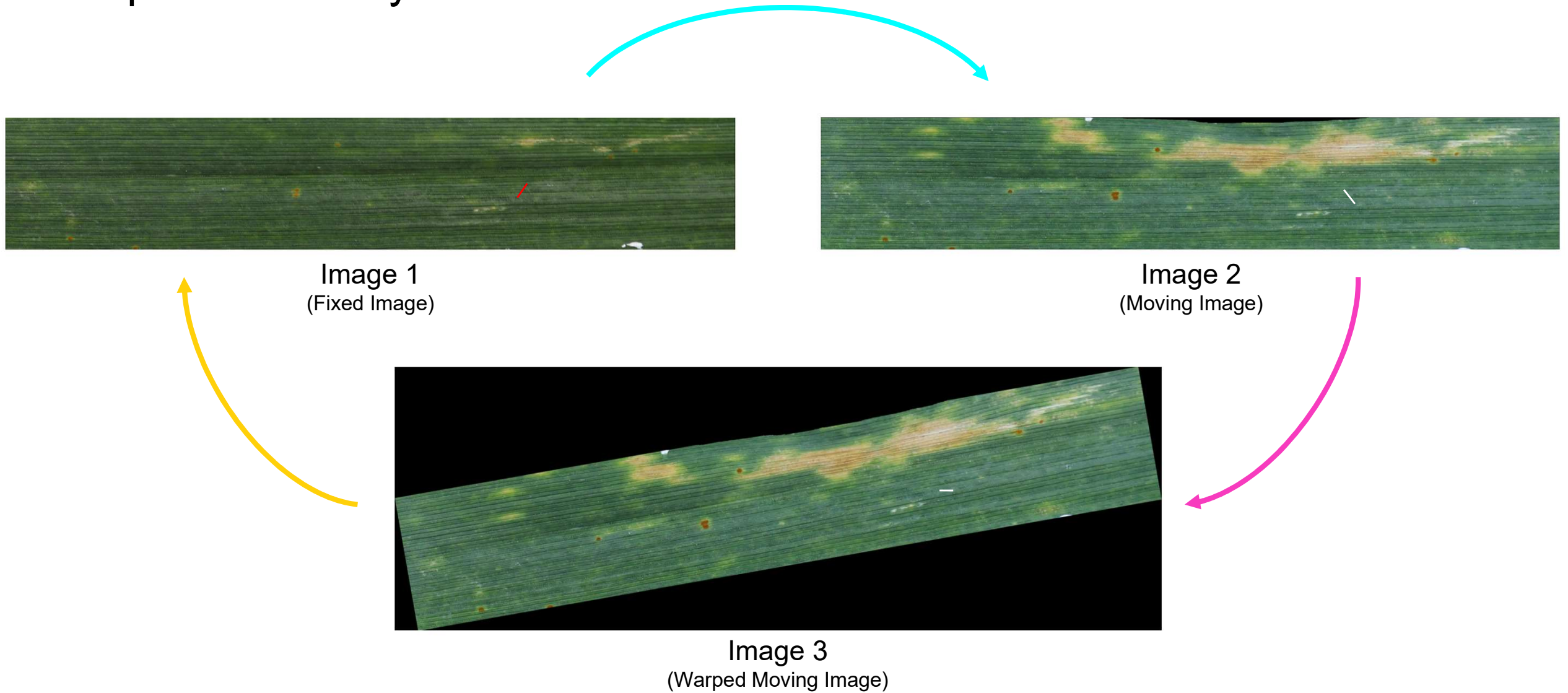


Method: LoFTR

- Dense feature matching method
- Can produce accurate dense matches even in low-texture regions
- **Method:**
 - extract dense matches at a coarse resolution
 - incorporate positional information and global context through transformers
 - refine coarse matches to a finer resolution
- **Output:**
 - Dense correspondences between image pairs
 - Confidence score for each correspondence
- **Issue**
 - Not all matches are accurate
 - Incorrect landmarks destabilize TPS



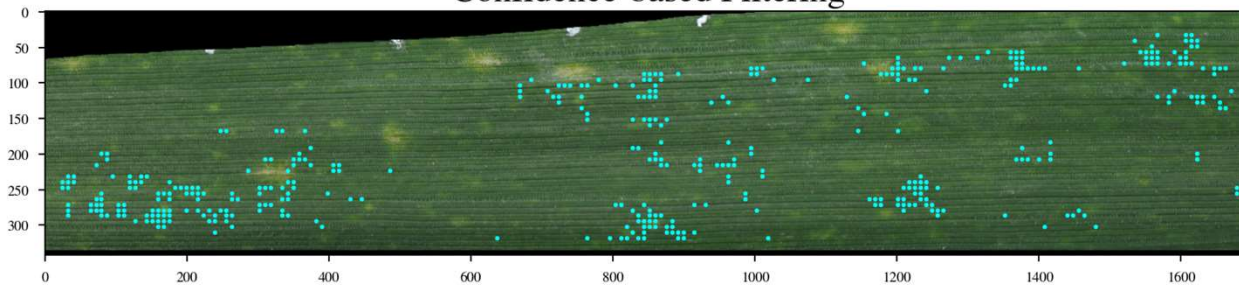
Method: Warp Consistency



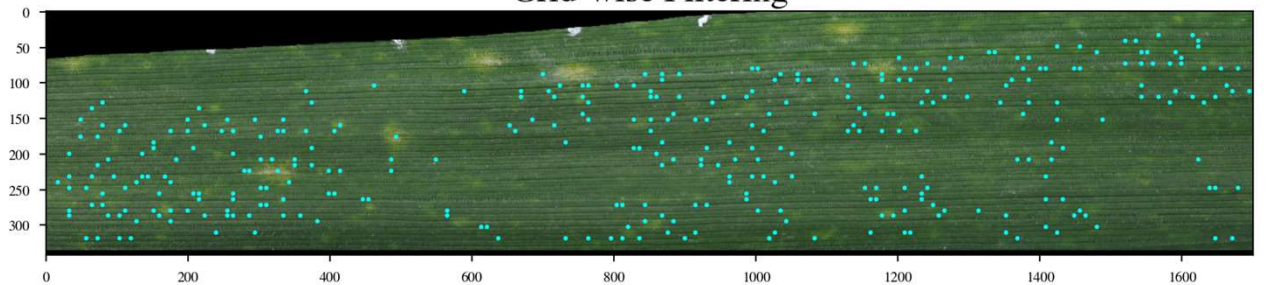
Method: Filtering Matches

- With over 500 keypoints, run into memory constraints when fitting the Thin Plate Spline
⇒ Use only a subset
- Minimum confidence threshold of 0.5
- Four potential filtering strategies:

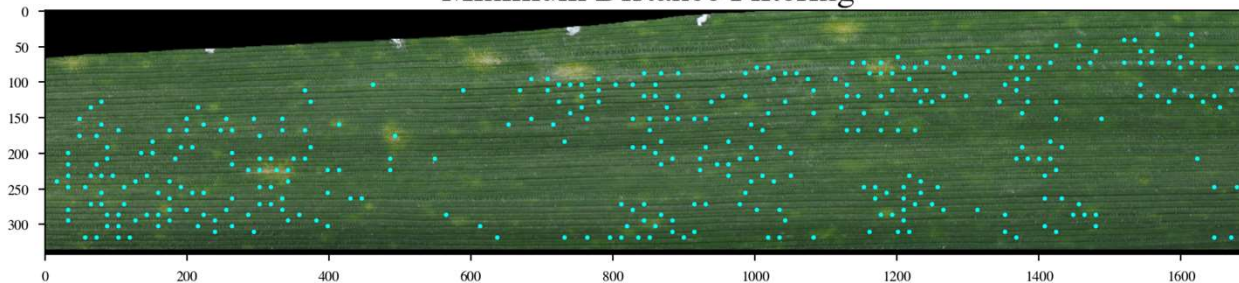
Confidence-based Filtering



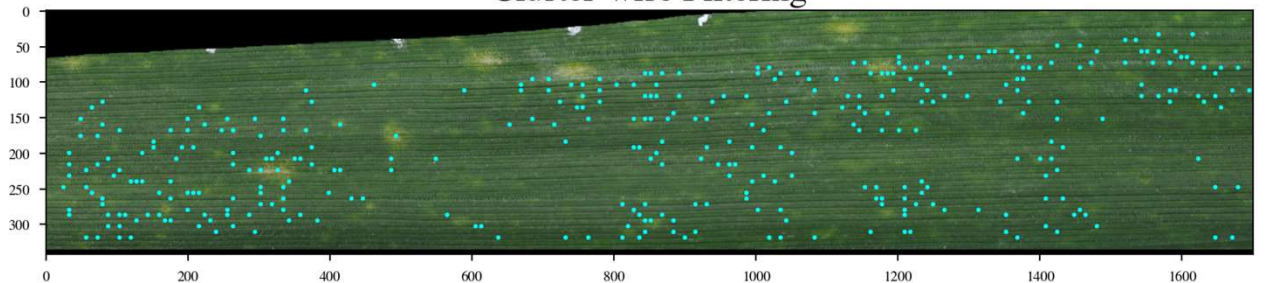
Grid-wise Filtering



Minimum Distance Filtering



Cluster-wise Filtering



Method:

Thin Plate Splines

- Classic transformation model in landmark-based registration for non-rigid deformations
 - Commonly used for medical images
 - Suitable for organic shape deformations due to penalization of high curvature transformations
- Uses arbitrary number of irregularly spaced corresponding landmarks in fixed and moving image
- Minimizes bending energy functional:

$$\min_f \sum_{i=1}^n \|p_i - f(q_i)\|^2 + \lambda \iint \left[\left(\frac{\partial^2 f}{\partial x^2} \right)^2 + 2 \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 + \left(\frac{\partial^2 f}{\partial y^2} \right)^2 \right] dx dy,$$

with solution

$$f(q) = P(q) + \sum_{i=1}^n w_i U(\|q - q_i\|),$$

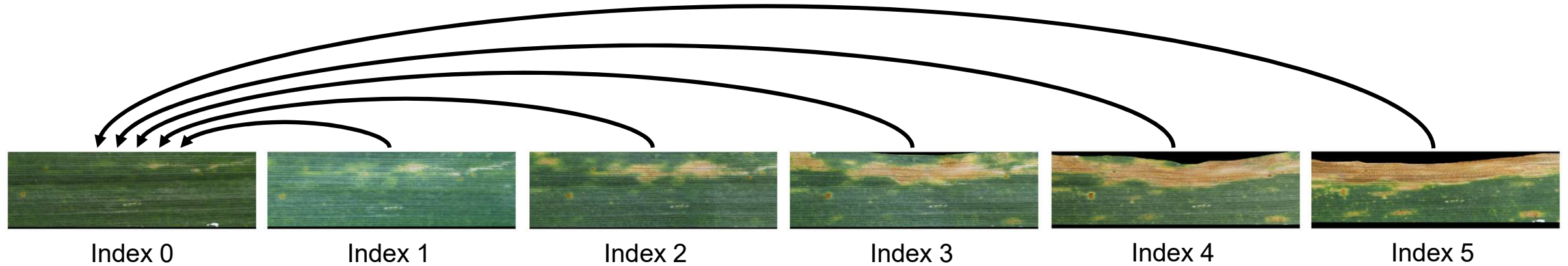
for P a first order polynomial, U radial basis function $U(r) = r^2 \log(r)$

- Closed-form solution exists
- Sensitive to incorrect or noisy landmarks

For details, see [5].

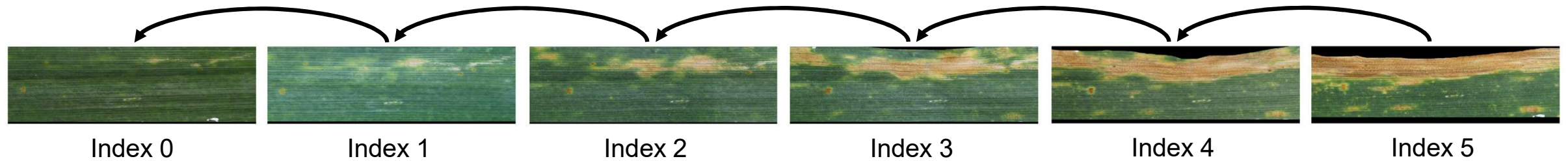
Method: Sequencing

- Leaf appearance can change drastically $\Rightarrow$ only few and low quality LoFTR matches
- **Question:** To which preceding image should we register?
- Three sequencing styles:
 - **Individual:** Register n-th image to first image



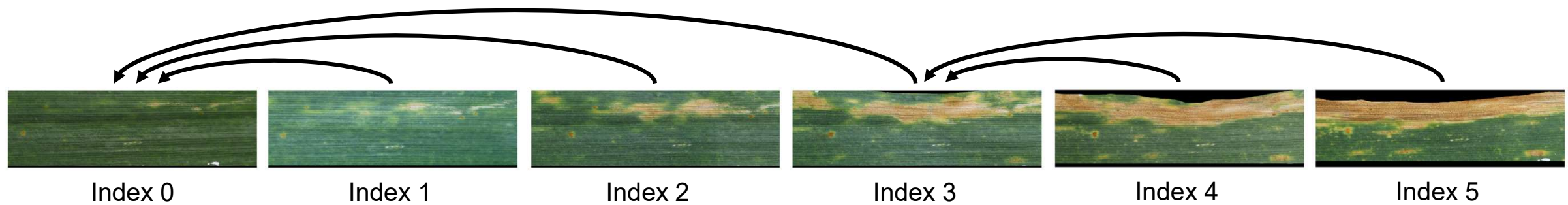
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Method: Sequencing

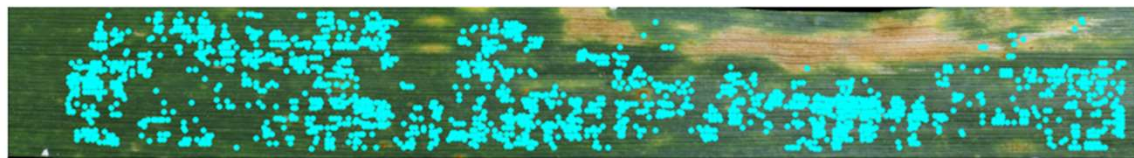
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- **Question:** To which preceding image should we register?
- Three sequencing styles:
 - **Individual:** Register n-th image to first image
 - **Sequential:** Register n-th image to (n-1)-th image and concatenate transformations
 - **Semi-Sequential:** Try to register n-th image to last "anchor" image and evaluate quality based on a criterion. If quality is insufficient, set (n-1)-th image as an "anchor" and register to that image.



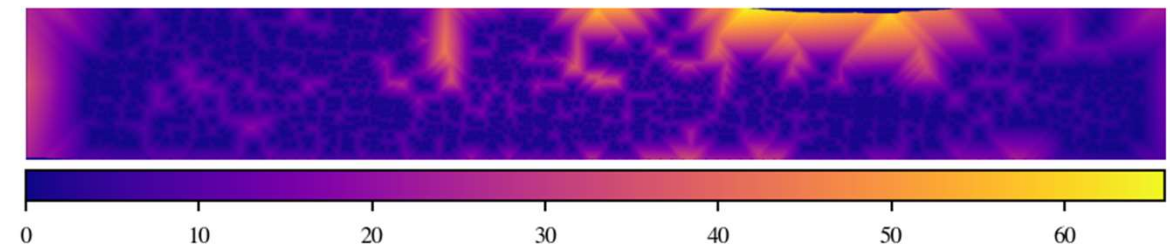
Method:

Criterion for Registration Quality

- **Goal:** Detect poor transformations
- Observation: large regions without matches often coincide with generally low-quality matches
⇒ Criterion based on distribution of matches
- For each pixel, compute distances to nearest keypoint
- Edges of the leaf are weighted less than interior of leaf



Match Distribution



Weighted distance to nearest keypoint

- Issue: just a proxy
 - Doesn't actually look at the quality of the registration/transformation

Method:

Evaluation & Metrics

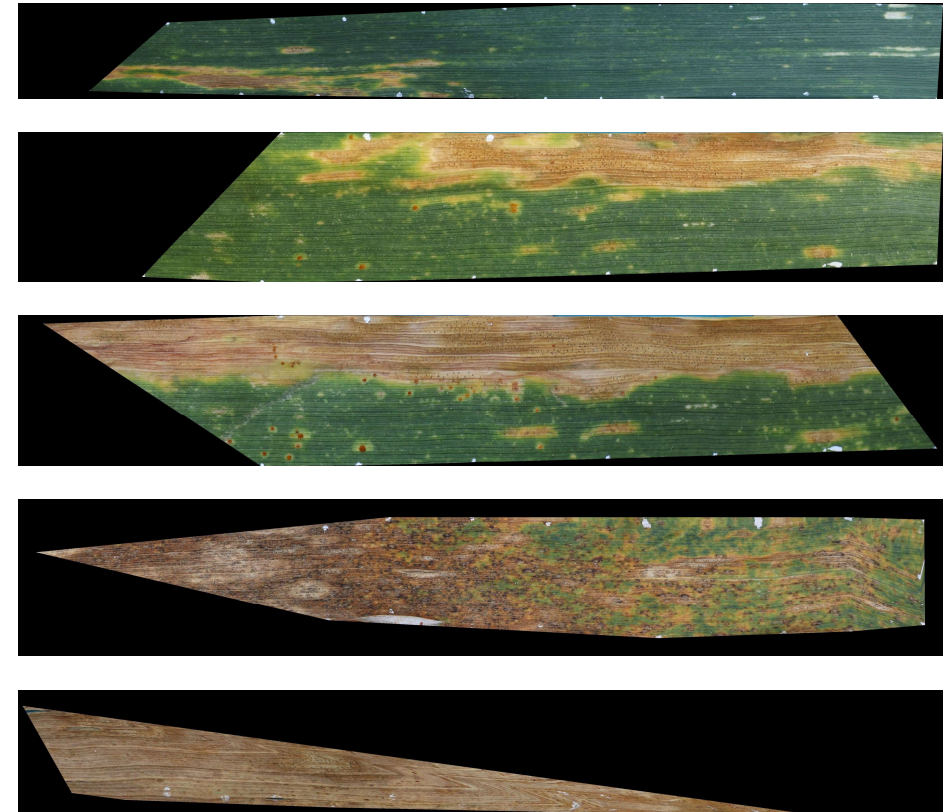
- **Evaluation Challenges:**
 - No ground truth (neither registrations nor landmarks)
 - New and growing lesions
- **Metrics:**
 - Referenced a selection of image registration survey and selected applicable metrics
 - Intensity-based Metrics:
 - **Mean Squared Error (MSE)**: very sensitive to lighting changes and growing lesions
 - **(Local) Normalized Cross Correlation (NCC)**: Robust to variations in brightness and contrast
 - **Normalized Mutual Information (NMI)**: invariant to monotonic transformations of the image intensities, but can be sensitive to noise and intensity variations.
 - **Structural Similarity Index (SSIM)**: constructed to focus on structural similarity between images
 - Mask-based Metrics:
 - **Intersection over Union (IoU)**: measure of overlap between two sets
 - **Hausdorff Distance**: measures the "worst case" misalignment of the two masks

Results:

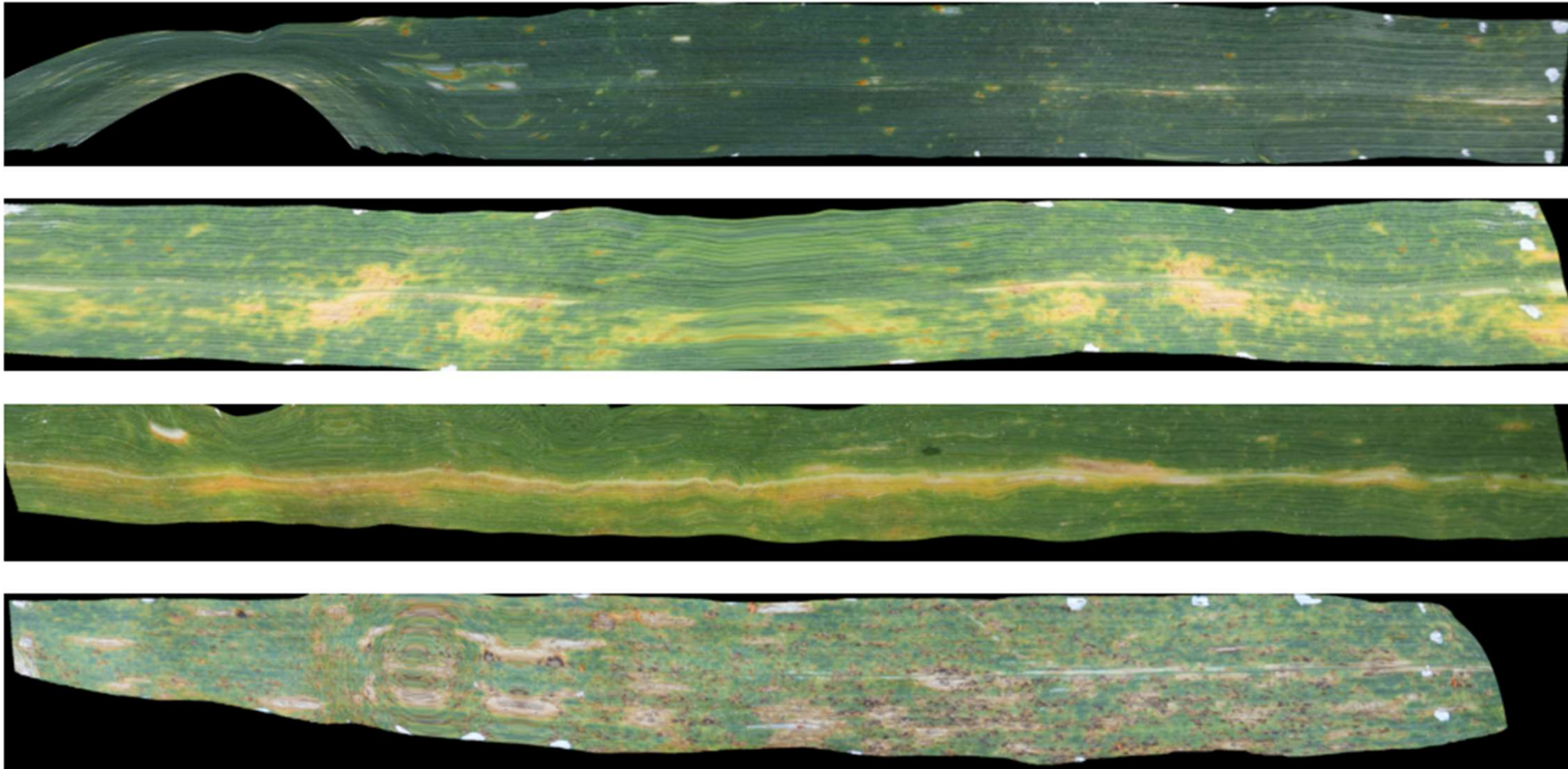
Comparison to Piecewise Affine Approach

- Both methods fail frequently on difficult late-state leaves
 - ⇒ Large appearance changes remain challenging
- Proposed method:
 - Works on full leaf
 - Performs well when changes to leaf are limited
 - Can become unstable
 - ⇒ Artifacts, erratic transformations, etc.
- Piecewise affine:
 - More stable
 - But loses leaf area due to missing markers

Piecewise Affine Failure Cases



Results:
Artifacts – Examples

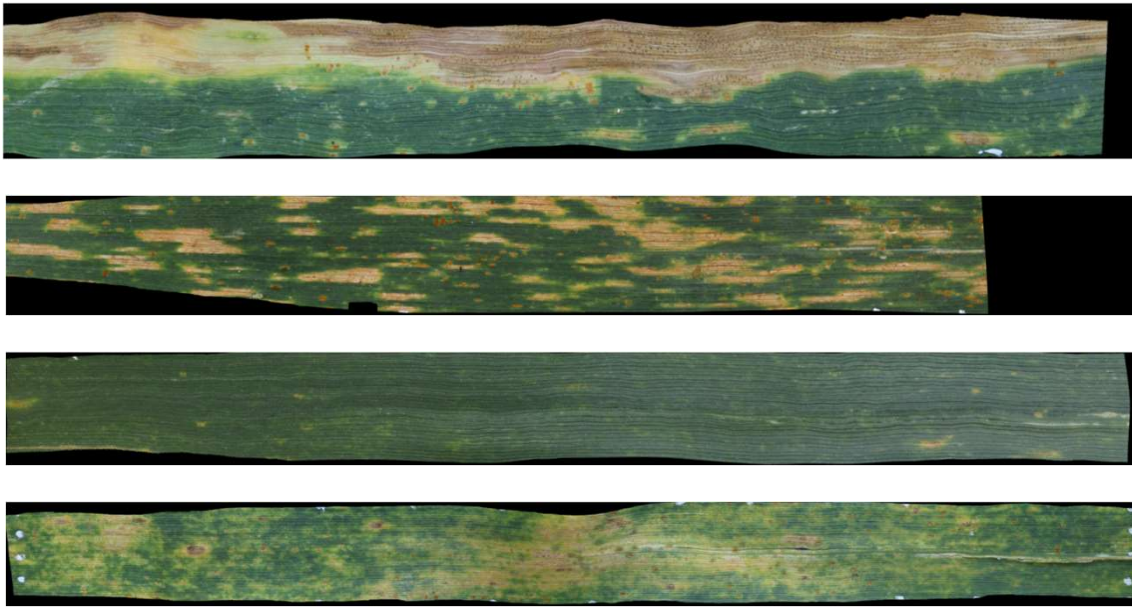


Results:

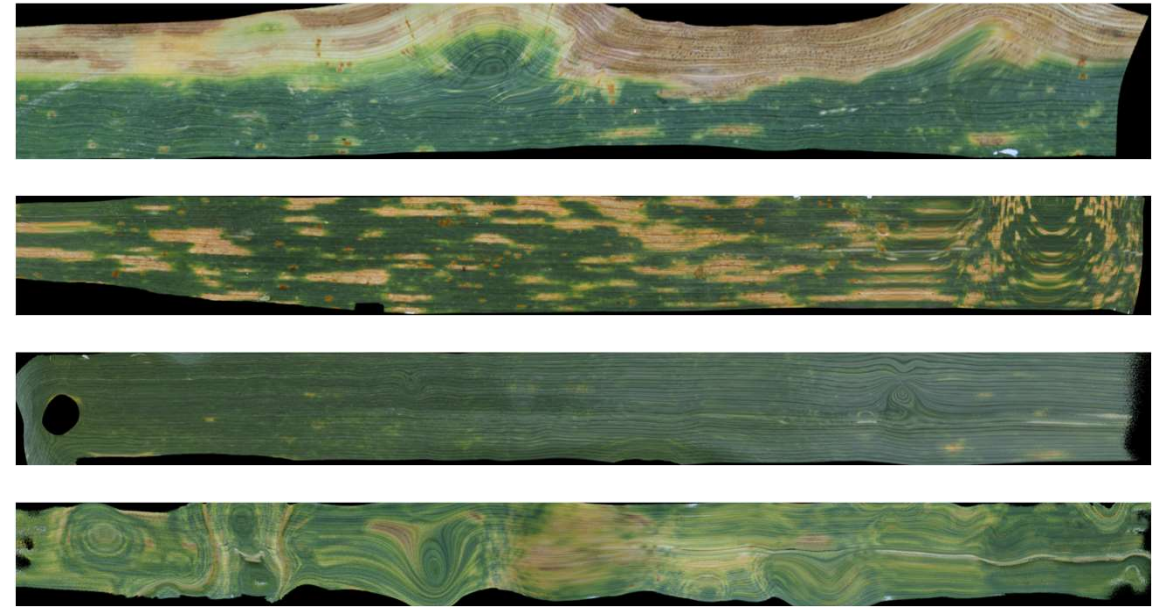
Warp Consistency is Crucial

- Without warp consistency:
 - Severe artifacts in most cases
 - Extremely unstable transformations
- Surprisingly not at all reflected in metrics

With Warp Consistency



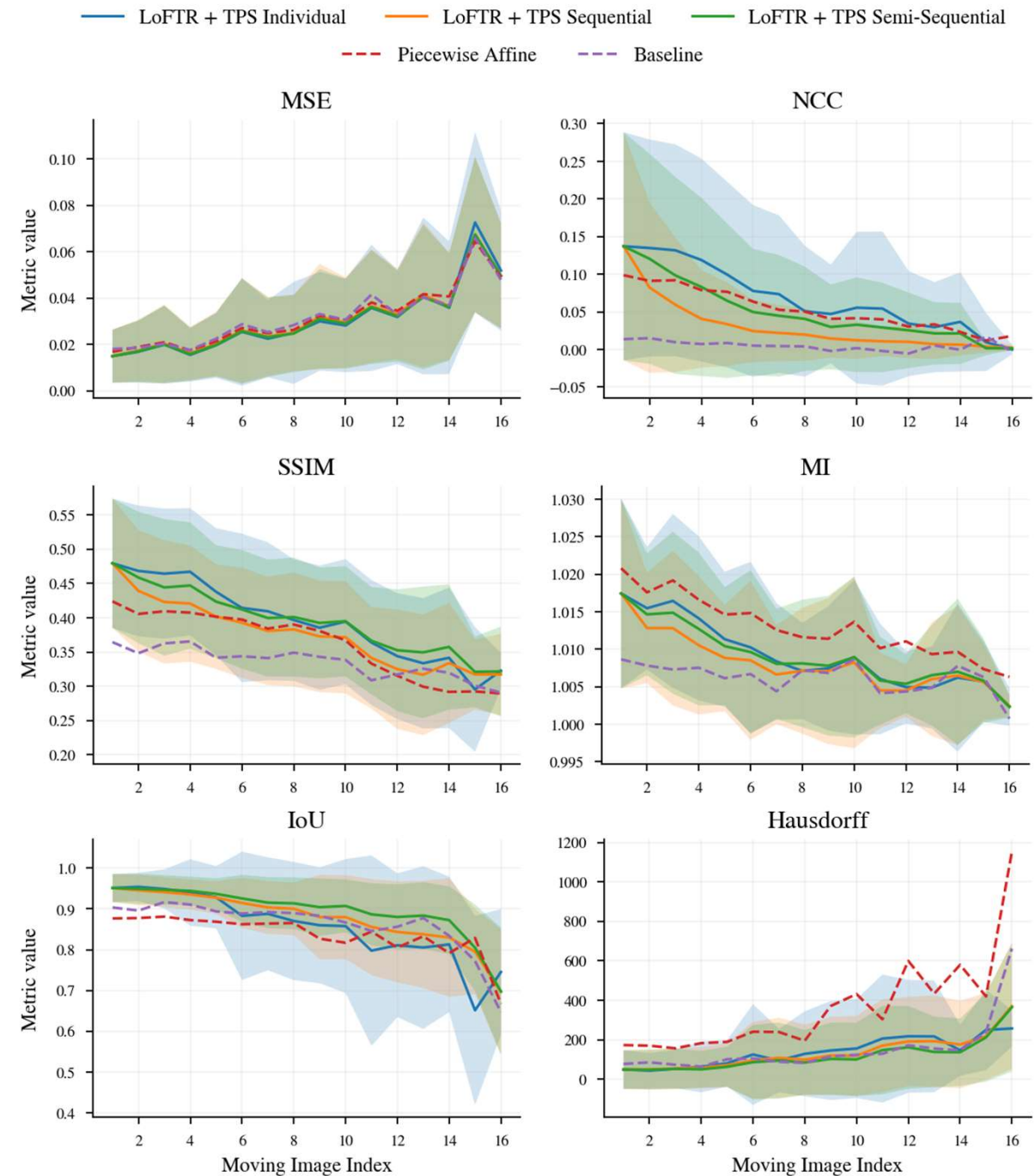
Without Warp Consistency



Results:

Comparing Sequencing Styles

- Individual
 - No error propagation
 - Good early performance
 - Highly unstable later
- Sequential
 - Less prone to complete deterioration than individual registration
 - Accumulates errors over time
- Semi-sequential
 - Best overall stability
 - Fewer severe artifacts



Discussion

- **Current limitations**

- Registration strongly depends on landmark quality
- Incorrect matches can cause artifacts, exaggerated deformations, and unstable TPS transforms
- Current evaluation metrics often fail to reflect visual quality

- **Future improvements**

- Better verification of LoFTR matches
- More robust / faster matchers
- Automatic detection of unstable registrations
 - E.g. TPS bending energy or Jacobian-based smoothness measures
- Improved evaluation metrics for non-rigid registration
 - E.g. segmentation-based metric that takes into account growth of lesions

Conclusion

- **Contributions**
 - Marker-free non-rigid registration pipeline
 - Full leaf area can be analyzed
 - Warp consistency + TPS regularization are crucial
 - Semi-sequential registration performs best
- **Limitations**
 - Sensitive to incorrect matches
 - Unstable for drastic appearance changes
 - No reliable quantitative assessment

Thank you for your attention!

Any Questions?

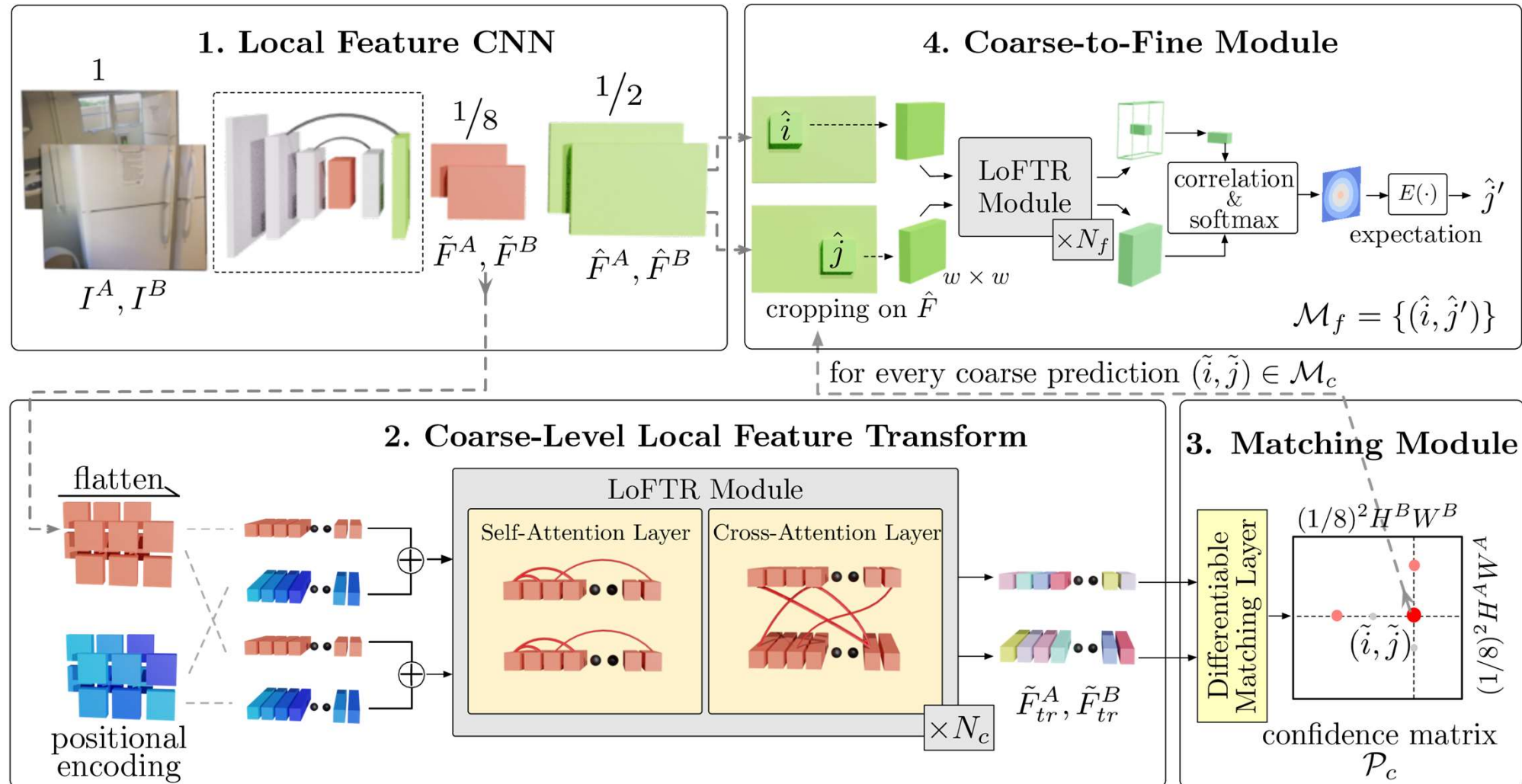
References

- [1] Jonas Anderegg, Radek Zenkl, Norbert Kirchgessner, Andreas Hund, Achim Walter, and Bruce A. McDonald. SYMPATHIQUE: Image-based tracking of symptoms and monitoring of pathogenesis to decompose quantitative disease resistance in the field. *Plant Methods*, 20(1):170, November 2024
- [2] Jonas Anderegg and Bruce A. McDonald. High-Resolution Leaf Image Sequences with Geometric Alignment for Dynamic Phenotyping of Foliar Diseases. *Scientific Data*, 13(1):247, January 2026.
- [3] Jonas Anderegg. High-Resolution Leaf Image Sequences with Geometric Alignment for Dynamic Trait Extraction. December 2025.
- [4] Jiaming Sun, Zehong Shen, Yuang Wang, Hujun Bao, and Xiaowei Zhou. LoFTR: Detector-free local feature matching with transformers. *CVPR*, 2021.
- [5] F.L. Bookstein. Principal warps: Thin-plate splines and the decomposition of deformations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 11(6):567–585, June 1989.

Supplemental Material

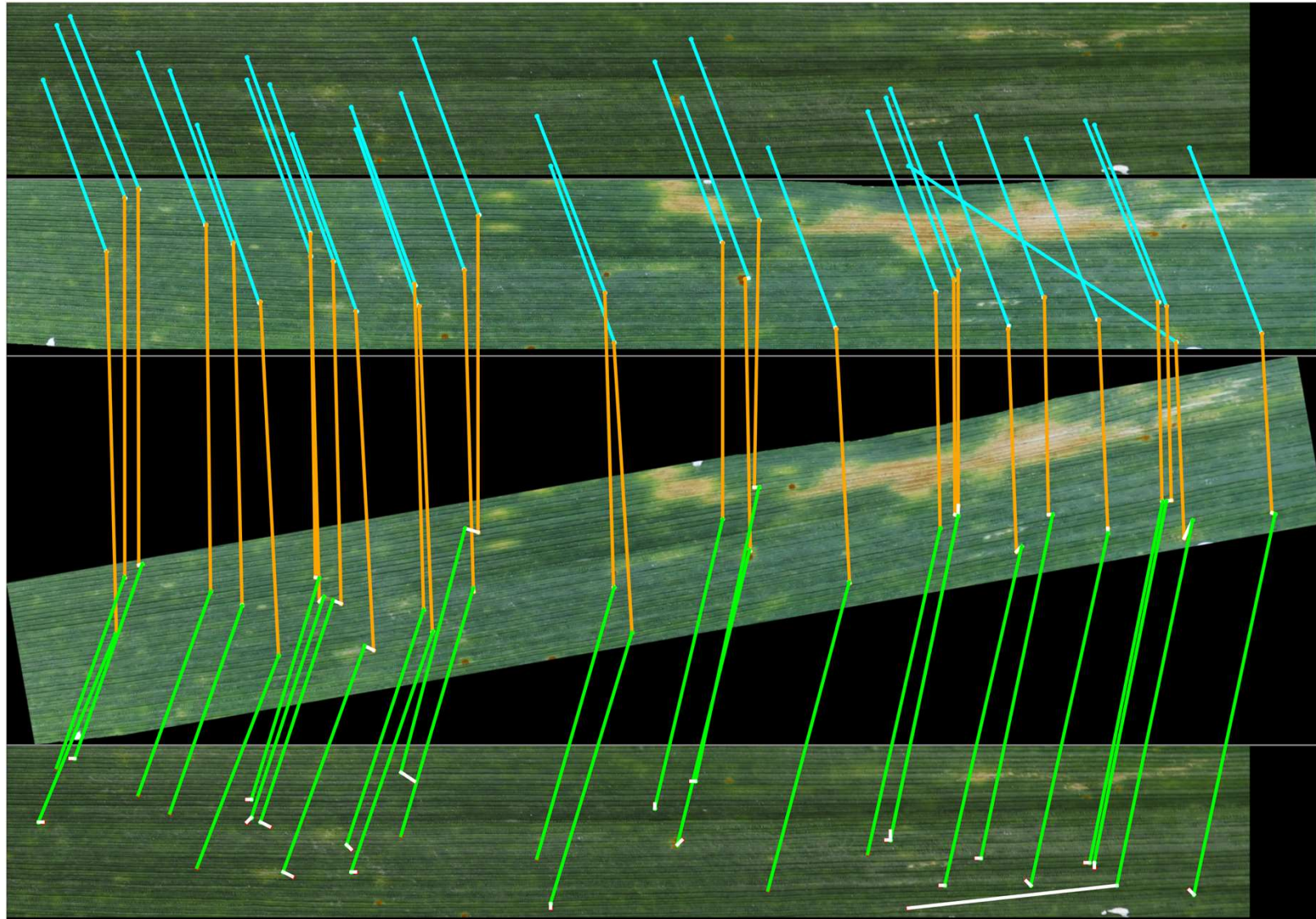
Method:

LoFTR – Details



Source: [4]

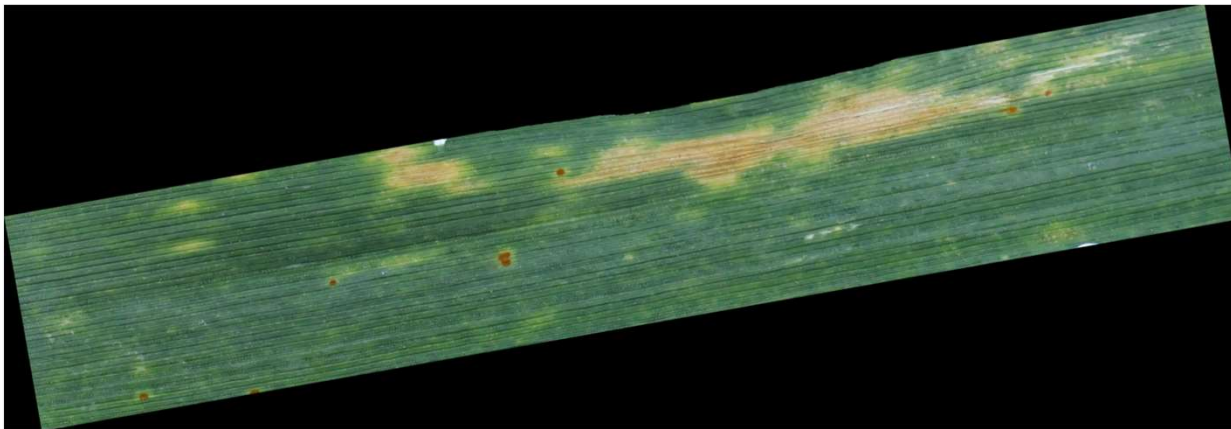
Method:
Warp Consistency Cycles



Method:

Warp Consistency – Choice of Transformation

- If transformation is too drastic, LoFTR struggles to find matches
 - Testing LoFTR robustness rather than match consistency
- Homography implies changes in scale and resolution
- Random transformation would introduce unnecessary uncertainty and inconsistent behavior



Rotation



Random Homography

Method:

Discarded Attempts for Quality Criterion

- **Goal:** Detect poor transformations
- Warp consistency-like approach for TPS
 - Method: Apply a transform to matches and generate a third image $\Rightarrow$ compare warped images
 - Issue: TPS is invariant under translations and rotations
 $\Rightarrow$ Discarded
- Ransac-like approach for TPS
 - Method: Fit the TPS for different subsets of matches and check for differences
 - Issue: only few iterations due to computational cost $\Rightarrow$ no stable results
 $\Rightarrow$ Discarded
- Thresholding based on Hausdorff distance and IoU of masks
 - Issue: doesn't detect artifacts within the leaf, and is susceptible to inherent masking differences
 $\Rightarrow$ Discarded

Method:

Metrics – Details: MSE

- Directly compares pixel values => very sensitive to lighting changes and growing lesions

$$\text{MSE}(I, \hat{f}) = \frac{1}{|\Omega|} \sum_{p \in \Omega} \|I(p) - \hat{f}(p)\|^2 ,$$

for Ω the domain of valid pixels, and $p \in \Omega$ a pixel in the image.

Method:

Metrics – Details: NCC

- Computes normalized cross correlation for local patches around each pixel
- Robust to variations in brightness and contrast
- For two image patches $A \subset I$ and $B \subset \hat{J}$, with means μ_A resp. μ_B and variances σ_A^2 resp. σ_B^2 , the local NCC is defined as

$$\text{NCC}_{\text{local}}(A, B) = \sum_p \frac{(A - \mu_A)(B - \mu_B)}{\sigma_A \sigma_B}.$$

- The local NCC is then averaged over the image

$$\text{NCC}(I, \hat{J}) = \frac{1}{|\Omega|} \sum_{p \in \Omega} \text{NCC}_{\text{local}}(I_p, \hat{J}_p)$$

- where $I_p, \hat{J}_p$ are the respective patches of I and $\hat{J}$ around pixel p .

Method:

Metrics – Details: NMI

- Models the probabilistic relation between the pixel intensities of the images.
- Primary metric used for multi-modal image registration in medical image registration
- Measures the amount of information that is shared between the two images.
- Invariant to monotonic transformations of the image intensities, but it can be sensitive to noise and intensity variations

- The NMI is defined as
$$\text{NMI}(I, \hat{I}) = \frac{H(I) + H(\hat{I})}{\mathcal{I}(I, \hat{I})},$$

•

where H is the marginal entropy
$$H(I) = - \int_{\mathbb{R}} p_I(x) \log(p_I(x)) \, dx,$$

and $\mathcal{I}$ is the mutual information
$$\mathcal{I}(x, y) = - \int_{\mathbb{R}} p_{I\hat{I}}(x, y) \log(p_{I\hat{I}}(x, y)) \, dx \, dy,$$

for joint probability $p_{I\hat{I}}$ and marginals p_I and $p_{\hat{I}}$

- The intensity distributions of both images are approximated with continuous histograms using Parzen-window estimates

Method:

Metrics – Details: SSIM

- Constructed to focus on structural similarity between images
- Composed of three components: luminance, contrast and structure:

$$\text{SSIM}(I, \hat{J}) = l(I, \hat{J})^{\alpha} \cdot c(I, \hat{J})^{\beta} \cdot s(I, \hat{J})^{\gamma}.$$

- The luminance comparison l is a function of the mean intensities of the images,
- The contrast comparison c operates on the standard deviations of the images.
- The structure comparison s is conducted on the normalized images
- The exponents $\alpha, \beta, \gamma > 0$ determine the importance of the components. In practice, they are commonly set to $\alpha = \beta = \gamma = 1$

Method:

Metrics – Details: IoU

- Normalized measure of overlap between two sets
- Denoting by $M_I, M_{\hat{J}}$ the respective masks of image I and $\hat{J}$, it is defined as:

$$\text{IoU}(I, \hat{J}) = \frac{|M_I \cap M_{\hat{J}}|}{|M_I \cup M_{\hat{J}}|}.$$

Method:

Metrics – Details: Hausdorff Distance

- Measures how far two sets are from each other
- Intuitively, measures the "worst case" misalignment of the two masks
- Defined as:

$$\text{Hausdorff}(I, \hat{J}) = \max \left\{ \sup_{p \in M_I} d(p, M_{\hat{J}}), \sup_{q \in M_{\hat{J}}} d(q, M_I) \right\},$$

where $d(q, M_I) = \inf_{p \in M_I} d(p, q)$ is the distance of point q to the set M_I

- In practice, one usually uses the 95th percentile of the Hausdorff distance to make the metric more robust to outliers.

Results:

Sequencing Comparison

Registration Method	Smoothing	MSE	NCC	MI	SSIM	IoU	Hausdorff
Baseline	-	0.0264	0.0062	1.0068	0.3451	0.8871	102.75
Piecewise Affine	-	0.0258	0.0645	1.0146	0.3850	0.8565	264.00
Individual	0	0.0238	0.0650	1.0099	0.3990	0.8897	106.71
	100	0.0238	0.0874	1.0110	0.4217	0.8949	106.67
Sequential	0	0.0254	0.0190	1.0070	0.3108	0.8110	129.57
	100	0.0246	0.0414	1.0095	0.3972	0.9079	96.13
Semi-Sequential	0	0.0250	0.0409	1.0085	0.3508	0.8555	108.33
	100	0.0242	0.0655	1.0107	0.4165	0.9221	84.22

Results:

Configuration Comparison

Configuration	MSE	NCC	MI	SSIM	IoU	Hausdorff
Default ($s = 100$)	0.0230	0.0766	1.0105	0.4131	0.9251	77.22
No Consistency	0.0228	0.0980	1.0112	0.4199	0.9341	56.31
Min. Dist. Filter	0.0228	0.1067	1.0117	0.4285	0.9319	65.42
Cluster Filter	0.0228	0.1047	1.0116	0.4270	0.9319	64.95
Grid Filter	0.0228	0.1066	1.0116	0.4290	0.9316	65.13
With Markers	0.0250	0.0606	1.0135	0.4283	0.9392	66.90
Full Leaf	0.0250	0.0433	1.0090	0.4618	0.8721	142.84
Unrotated ROI	0.0225	0.0707	1.0108	0.6450	0.9075	66.23
Unrotated ROI (Homography)	0.0229	0.0615	1.0104	0.6344	0.8972	70.03

Discussion:

LoFTR Alternatives

- EfficientLoFTR
 - Same strengths as LoFTR, but faster inference
 - Could e.g. enable repeated iterations of Warp Consistency
- SuperPoint + LightGlue
 - Popular LoFTR alternative: robust and accurate
 - Can struggle in low-texture cases

Results:

Further Qualitative Findings

- Including markers yields only marginal increase in robustness.
- Filtering styles lead to no significant difference.
 - For 500 matches, leaf is covered sufficiently well for majority of cases.
- Regularization improves stability drastically.
- Coverage-based criterion is often too quick to reset anchors.